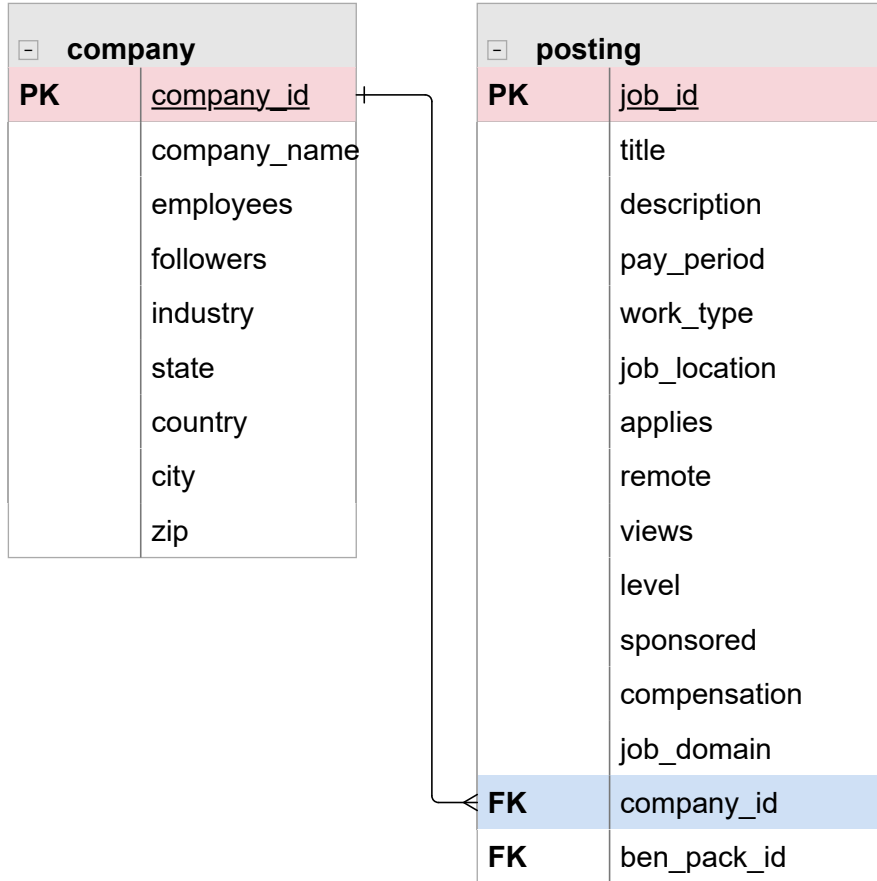




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ERD: Primary / Foreign Keys

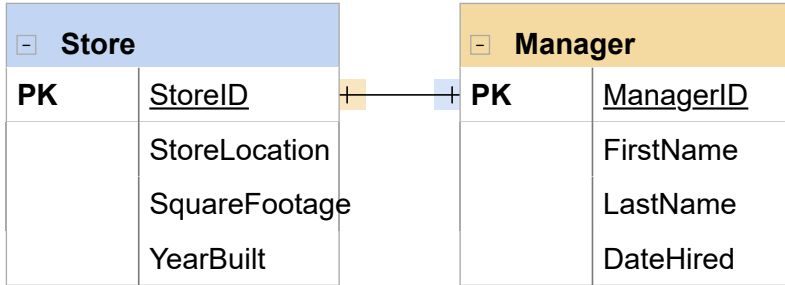


A **primary key** is a unique identifier (think social security number)

A **foreign key** is another table's primary key, used to link two tables together

Entity
Relationship
Diagram

ERD: Cardinality

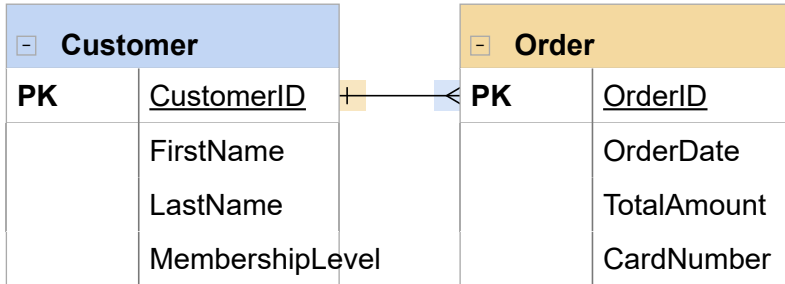


Read left-to-right AND right-to-left

One to One (1:1)

A store has one manager

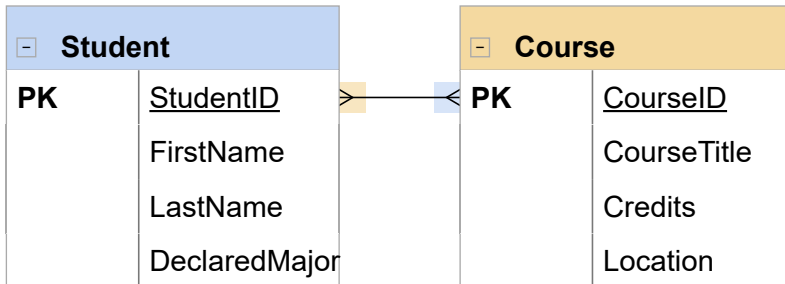
A manager works at one store



One to Many (1:N)

A customer can have multiple orders

An order belongs to one customer



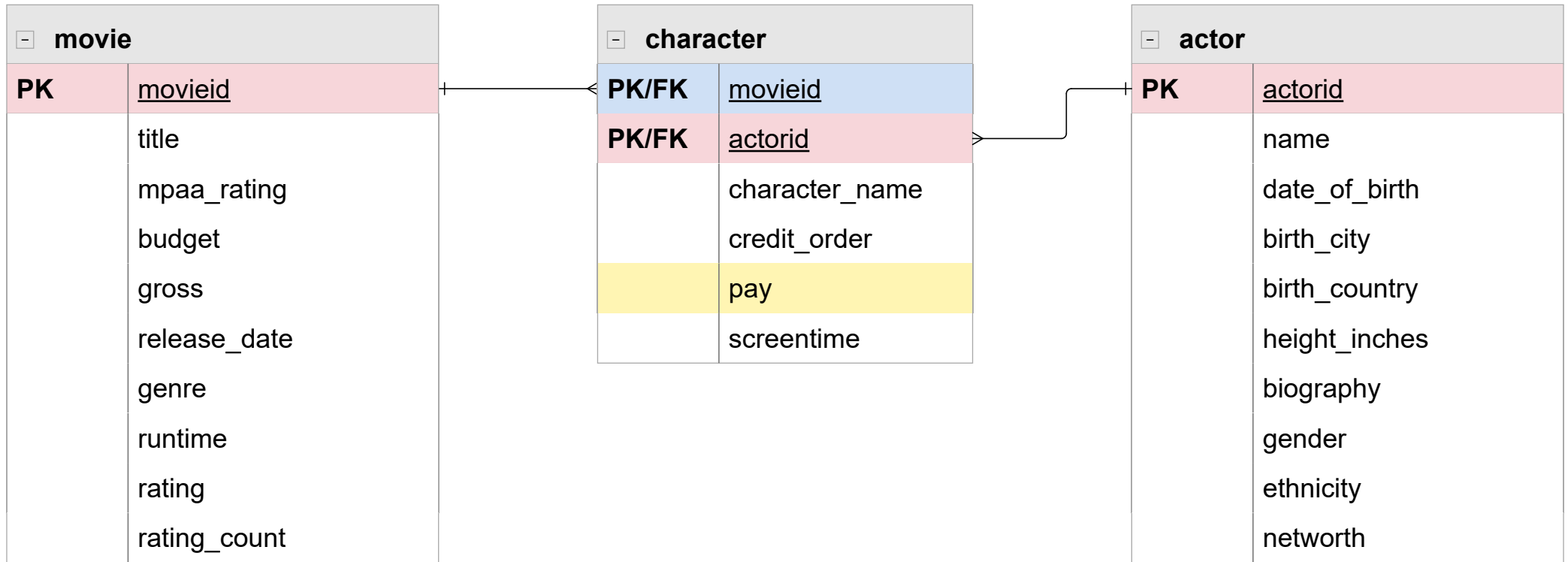
Many to Many (M:N)

A student can enroll in many courses

A course can have lots of students

Note! M:N cardinality requires a composite table (next slide)

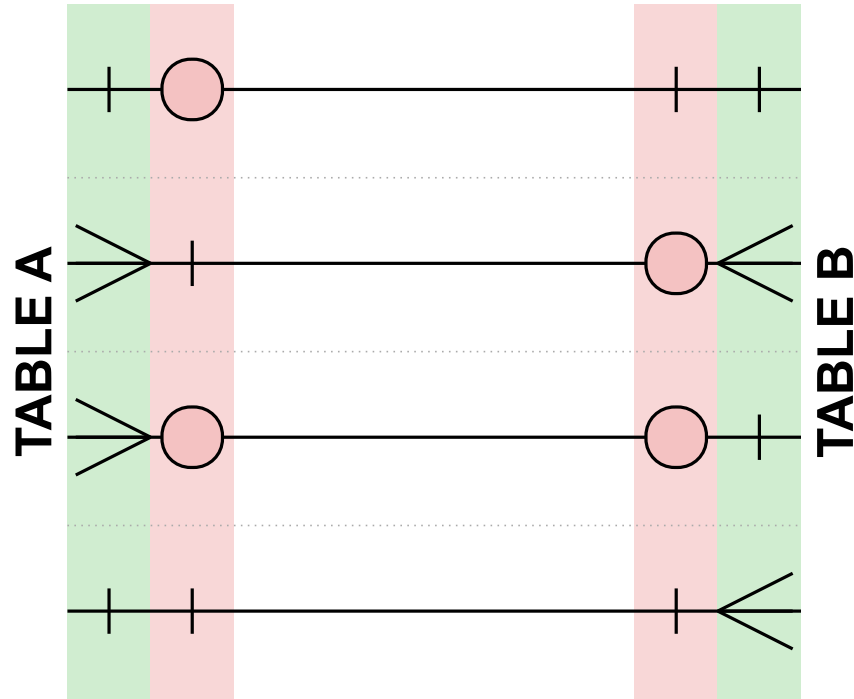
ERD: Composite Table



To know how much money Tom Hanks got **paid** to play Woody in *Toy Story*, you need to know both the **movie** and the **actor**. His pay is probably different than when he was Woody in *Toy Story 2*, or when he was Forrest Gump in *Forrest Gump*

ERD: Minimum / Maximum Cardinality

The inner marks are minimum (0 or 1)
and the outer marks are maximum (1 or many)



A could have 1 B or 1 B

B could have 0 A or 1 A

A could have 0 B or many B

B could have 1 A or many A

A could have 0 B or 1 B

B could have 0 A or many A

A could have 1 B or many B

B could have 1 A or 1 A

Think: a student could have 0 cars, but they could also have multiple.

SQL

```
1 SELECT      -- attributes -- do SELECT DISTINCT to only show unique results
2 FROM        -- tableA
3 JOIN        -- tableB ON tableA.attribute = tableB.attribute (order doesn't matter)
4 WHERE       -- attribute filters [=, !=, <>, IS, LIKE '%__%', IN ("__","__","__")]
5     AND/OR  -- you only say WHERE once, but can have many filters
6 GROUP BY    -- all non-aggregated attributes when aggregating
7 HAVING      -- group-based filters (not used in the SQL project)
8 ORDER BY    -- attributes in ASC (default) / DESC order
9 LIMIT       -- to xx top results
10
11 -- aggregate(attribute) AS newName
12 count()
13 avg()
14 min()
15 max()
16 -- etc.
```

Structured
Query
Language

SQL: Where Use Cases

ta_name = "James"

→ Exact match

ta_name != "James"
ta_name <> "James"

→ Not exact match
(these are equivalent)

ta_name **IS** NULL

→ NULL means blank data
(can also do **IS NOT** NULL)

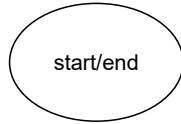
ta_name **LIKE** '%ame%'

→ The text ("*ame*") is contained within the attribute
(% = *wildcard*)

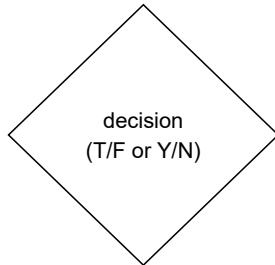
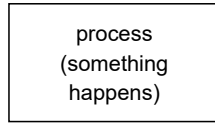
ta_name **IN** ("James", "Robert", "Frankie")

→ Exact match for any of these

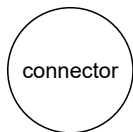
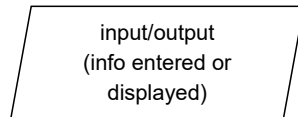
Flow Charts



you can only have ONE start
but you CAN have multiple
ends



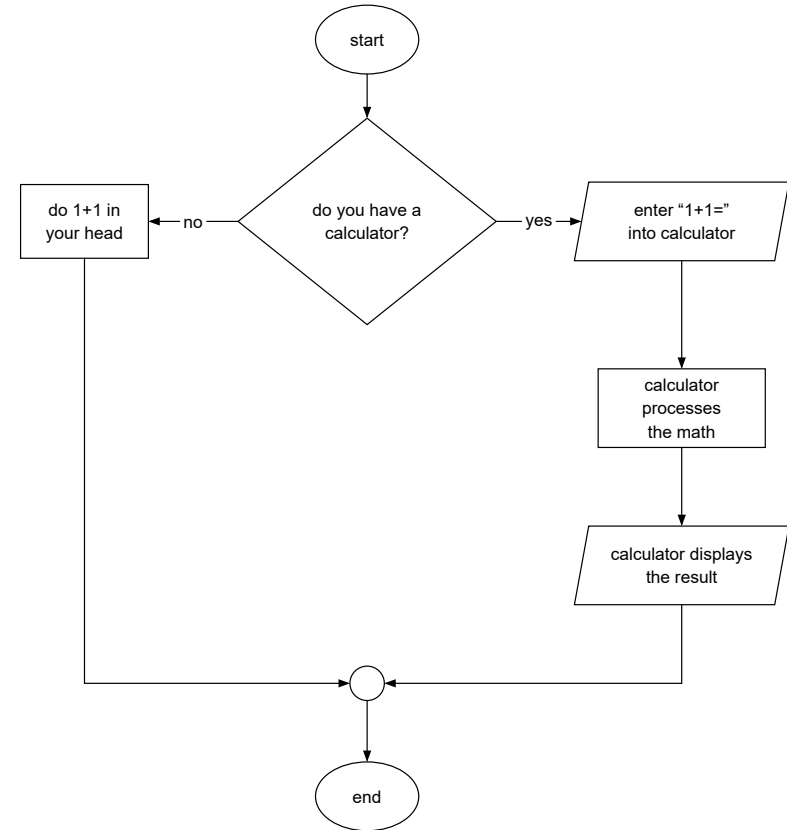
decisions are the ONLY thing
that can have more than one
arrow pointing OUT of them



connectors are the ONLY
thing that can have more than
one arrow pointing INTO them

file > export as > pdf

EXAMPLE: How to solve 1+1



VBA: Basics

Option Explicit 'This makes it so that you can only use variables you've declared (VERY RECOMMENDED)

Sub thisIsMySubName()

 'BASICS!

 'Generally you will first refer to an object type (like a Sheet)

 'and then tell it what you want to do (like delete or add or copy, etc):

 ' Sheets and Worksheets are the same thing

 Sheets("MySheet").Delete

 Sheets.Add.name = "MySheet"

 Worksheets("MySheet").Activate

 ' Range and Cells are SIMILAR but different: Range("E7") = Cells(7, 5)

 Range("A1").Activate

 ActiveCell.Value = "Hello"

 Range("A1:D3").Copy

 Range("E5").PasteSpecial

 ' Columns and Rows are what they sound like

 Columns("B:D").Delete

Visual
Basic for
Applications

VBA: Navigation and Misc.

'NAVIGATION!

```
'
ActiveCell.End (xlDown)
ActiveCell.End (xlToLeft)

' Move (rows, columns) based on a cell or range
ActiveCell.Offset(1, 2).Value = "NewSpot"
    Range("A1").Offset(3, 4).Value = "NewSpot" 'The cell that now says "NewSpot" is E4
```

' MISCELLANEOUS!

```
' Bolding
Range("A1").Font.Bold = True

' Autofitting columns
Columns.AutoFit 'all columns
Columns("A:C").AutoFit 'specific columns

' Currency format
Range("A1").Style = "Currency"

' Bottom border
Range("A1").Borders(xlEdgeBottom).LineStyle = xlContinuous

' Text splicing
name = "James"
first_initial = Left(name, 1)
```

VBA: Variables

```
'DECLARE VARIABLES!
Dim i As Integer    'a whole number
Dim j As Double     'a number with a decimal
Dim k As String     'a line of text
Dim l As Boolean    'true or false
Dim m As Worksheet

i = 5
j = 1.2345
k = "Hello there! General Kenobi!"
l = True
Set m = Sheets("Sheet1") 'objects (sheets, books, ranges) use Set instead of =

'USER INTERFACE!

'Tell the user something
MsgBox "Oh hello there user, welcome to my spreadsheet"

'Ask for an input
Dim username As String
username = InputBox("What's your name?")

' COMBINING STRINGS

Dim myOutput As String
Dim name As String
name = "James"

' COMbine strings with ampersands (&)-- don't forget spaces
myOutput = "Hello. My name is " & name & " and this is Disney Channel!"
```

'CONDITIONALS!

```
' If Statements
If i < 0 Then
    ' your code here
ElseIf i < 5 Then
    ' your code here
ElseIf i < 10 Then
    ' your code here
Else
    ' your code here
End If
```

```
'Case Statements
Select Case fruit
    Case "apple", "strawberry"
        Color = "red"
    Case "grape"
        Color = "purple"
    Case Else
        Color = "unknown"
End Select
```

```
'Another Case Statement
Select Case temperature
    Case Is >= 100
        feel = "too hot"
    Case 80 To 100
        feel = "pretty warm"
    Case Is < 80
        feel = "I'm probably staying out of the pool"
End Select
```

VBA: Conditionals

'LOOPS!

VBA: Loops

```
'For Loop
For i = 0 To 5
    ' your code here
Next i
```

```
'For Each Loop
Dim x As Range
For Each x In Range("A2", Cells(lastrow, "A"))
    ' your code here
Next
```

```
'For Each Loop 2
Dim x As Range
Dim mtRange As Range
Set myRange = Range("A1:E5")
For Each x In myRange
    ' your code here
Next
```

```
'Do While Loop
Do While i = 5 'condition does NOT have to be numbers
    ' your code here
Loop
```

```
' Do Until Loop
Do Until i = 10 'condition does NOT have to be numbers
    'your code here
Loop
```

End Sub

VBA: Functions

```
Function thisIsMyFunctionName(num As Integer, tf As Boolean) As String
```

```
' a function declaration is just like a set of dims  
' the INPUT is in the parentheses ()  
' the OUTPUT is outside of it  
' so this function KIND OF has this baked into it:
```

```
Dim num As Integer  
Dim tf As Boolean  
Dim thisIsMyFunctionName As String
```

```
' the INPUT then comes from your excel sheet  
' for example, in your sheet you cell might say:  
' =thisIsMyFunctionName(15, True)
```

```
' the OUTPUT comes from your function  
' set the function's name equal to something  
' in this case, our output has to be a string:
```

```
thisIsMyFunctionName = "That's a very nice hat you have there, partner!"
```

```
' generally your inputs will be used to calculate/decide your output
```

```
End Function
```

Statistics

Null Hypothesis

H_0 : No x's are significant

Alt Hypothesis

H_A : At least one x is significant

Scientific Notation:

$$aEb = a \times 10^b$$

$$aE3 = a \times 1000 \text{ (big)}$$

$$aE(-3) = a \times 0.001 \text{ (small)}$$

Test	Variables	Stat
Correlation	numeric – numeric	r
Regression	numeric – multi. numeric	R^2
ANOVA	numeric – mult. categorical	p
T-test	numeric – 2 pair categorical	p

r : -1 to 1 (as $x \uparrow$, $y \downarrow$ / $y \uparrow$)

R^2 : 0 to 1 (share of variance in y explained by x)

p : 0 to α (usually 0.05) $\rightarrow$ reject the null if

$$p < \alpha$$

Tableau

- “Default” settings are usually a good starting point — check them first
- Switching the x- and y-axis can reveal a clearer story
- Watch your sort/ordering — Tableau doesn’t always order the way you expect

Solver

Two important Excel formulas:

1 =SUM(A1:A5) → A1 + A2 + A3 + A4 + A5

2 =SUMPRODUCT(A1:A5, B11:B15) → A1*B11 + A2*B12 + ... + A5*B15

3 (the two ranges should be the same size and shape)

- **Constraints:** set a changing-cell reference, an operator ($\leq$, $=$, $\geq$), and the constraint value
- **If integer constraints aren't working:** Solver $\rightarrow$ Options $\rightarrow$ uncheck "Ignore Integer Constraints" and set Integer Optimality to 1%

HTML: Setup

In VS Code, inside a .html file, type ! then Enter to generate:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-
6      scale=1.0">
7      <title>Document</title>
8  </head>
9  <body>
10 </body>
11 </html>
```



startbootstrap.com/themes/portfolio-resume

HyperText
Markup
Language

HTML: Tags



```
1  <!-- This is a comment -->
2  <h1>Main heading</h1>
3  <h2>Section heading</h2>
4  <h3>Subsection</h3>
5  <p>This is a paragraph.</p>
6  <!-- Line break -->
7  <br>
8  <!-- Unordered list (bullets) -->
9  <ul>
10     <li>Item 1</li>
11     <li>Item 2</li>
12 </ul>
13 <!-- Ordered list (numbers) -->
14 <ol>
15     <li>Item 1</li>
16     <li>Item 2</li>
17 </ol>
```

The start and end tags act like parentheses: <tag> stuff </tag>

HTML: Anchors



```
1 <!-- Hyperlinks -->
2 <a href="https://www.google.com">Visit Google</a>
3 <!-- Folder references -->
4 <a href="otherpage.html">Visit Other Page</a>
5 <a href="subfolder/page.html">Visit Subfolder Page</a>
6 <a href="../yetanotherpage.html">Visit yet another page</a>
7 <!-- On-page anchors (any tag can have an id, not just <a>) -->
8 <a id="section1">Section 1</a>
9 <a href="#section1">Go to Section 1</a>
```

../ means go to the parent folder (or root directory)

HTML: Images



```
1 <!-- Images -->
2 
3 
```

Notice: images do NOT have an end tag `</img>` (neither do line breaks `</br>`)

HTML: Divisions

Divs don't inherently do anything — they're invisible boxes you put content into so you can isolate it for positioning and CSS styling using classes.

HTML

```
1 <div class="square-  
  image">  
2       
3 </div>
```



CSS

```
1 .square-image {  
2     width: 200px;  
3     height: 200px;  
4 }
```



For more, see: **CSS: Selectors/Properties/Values**

HTML: Embedding

Sometimes doesn't work until you go live

YouTube

1. Click **Share**
2. Click **Embed**
3. Click **Copy**
4. Paste into your code

Tableau

1. Click the share icon
2. Click **Copy Embed Code**
3. Paste into your code

HTML: Folder Structure & Formatting

Use folders! Keep `css/`, `html/`, `images/`, and `js/` separate, with `index.html` at the project root.

To quickly format everything correctly (tabs, long lines, etc.) in VS Code:

Right click → Format Document

HTML: Uploading to GitHub

1. Click **New** to create a repository
2. Click **Create repository**
3. On the repo page: **Add file** → **Upload files**
4. Click **Commit changes**

Upload the CONTENTS of your folder, not the folder itself

Going Live

1. Go to **Settings**
2. Click **Pages** in the sidebar
3. Under Branch, select `main` (instead of `None`) and **Save**

After a couple of minutes, refresh the page — there will be a link to your website at the top.

CSS: Basics

```
1 <!-- Connect CSS to HTML, in your <head> -->
2 <link rel="stylesheet" href="styles.css" />
```



```
1 selector {
2     property: value;
3     property2: value2;
4 }
```



CSS is **cascading** because more recent styling supersedes previous styling — if you set the font color to blue, then later set it to red, it will be red.

Cascading
Style
Sheets

CSS: Selectors / Properties / Values

```
1  /* Built-in selectors: the tags you use in your HTML (body, h1, p, a, img...) */
2  body {
3      background-color: #f0f0f0;
4      font-family: Arial, sans-serif;
5      margin: 0;
6      padding: 20px;
7  }
8  /* Class (custom) selectors: create your own with a period + name */
9  .highlight {
10     background-color: yellow;
11     font-weight: bold;
12 }
```

```
1 <p class="highlight">This text will be highlighted.</p>
```



There are A LOT of selectors and properties — Google and AI are your friend for finding exactly what you want!

jimna-h.github.io/james_super_cool_is_201_cheatsheet



IS 201 TA Lab